

Lecturer-Unit Relationship via Register Table

In modern university systems, it is common for lecturers to co-teach units or to be assigned to specific units for specific academic terms. To handle this dynamic relationship, the traditional one-to-many model (where each unit has one lecturer) is replaced with a many-to-many model using a junction table called 'Register'.

This Register table serves the same purpose as the Enrollment table used for students, allowing us to record not only which lecturers are teaching which units, but also during which semester and academic year. This enables co-teaching, teaching load tracking, and semester-specific assignments.

Structure of the Register Table:

```
Register (  
  register_id INT PRIMARY KEY AUTO_INCREMENT,  
  lecturer_id INT,  
  unit_id INT,  
  semester VARCHAR(10),  
  academic_year VARCHAR(10),  
  role VARCHAR(50), -- e.g., 'Lead Lecturer', 'Assistant'  
  FOREIGN KEY (lecturer_id) REFERENCES Lecturer(lecturer_id),  
  FOREIGN KEY (unit_id) REFERENCES Unit(unit_id)  
)
```

By removing lecturer_id from the Unit table and introducing the Register table, we achieve a more flexible and scalable structure. Now, a unit can be taught by multiple lecturers, and a lecturer can

teach different units in different semesters. This approach also supports the recording of lecturer roles (e.g., lead or assistant lecturer), which is essential for institutions tracking responsibilities and performance.

Benefits of Using the Register Table:

- Supports co-teaching and role distinctions
- Tracks teaching per semester and academic year
- Facilitates performance tracking and workload management
- Aligns with the normalized database design principles

This method reflects the realities of academic teaching environments in universities globally and improves the system's adaptability and extensibility.